BATCH NO:	23
PROJECT TITLE	AN EXPLAINABLE AI AGENT – BASED FRAMEWORK FOR MULTI-CLASS VIOLENCE DETECTION IN SURVEILLANCE VIDEOS

ABSTRACT

Violence detection in surveillance systems is crucial for improving public safety and enabling rapid emergency response. However, most existing deep learning approaches focus only on binary classification of violent and non-violent activities, limiting their ability to recognize different types of anomalies and provide interpretable results. These systems often struggle with false alarms, limited contextual understanding, and poor generalization. To address these issues, this paper proposes an explainable AI agent–based framework for multi-class violence detection in surveillance videos. The system first applies YOLOv8 for human-centric frame filtering to reduce unnecessary computation. A Video Swin Transformer is then used to extract spatiotemporal features and perform multi-class classification. A confidence-based decision mechanism is introduced to minimize false positives, and a Large Language Model (LLM) generates human-readable explanations to improve transparency and trust. The proposed framework identifies multiple anomaly categories, including fights, robbery, panic movement, and stampede. Experimental results demonstrate improved robustness and reliable multi-class performance compared to conventional CNN-based methods, highlighting the effectiveness of transformer-based explainable surveillance systems.

KEYWORDS: Violence detection, surveillance systems, multi-class classification, anomaly detection, explainable AI, YOLOv8, Video Swin Transformer, spatiotemporal features, confidence-based decision, large language model.

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